



RESPIRATORY EMERGENCIES

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AGENDA

Review Scenario 1

01

Review Scenario 2

02

Review Scenario 3

03

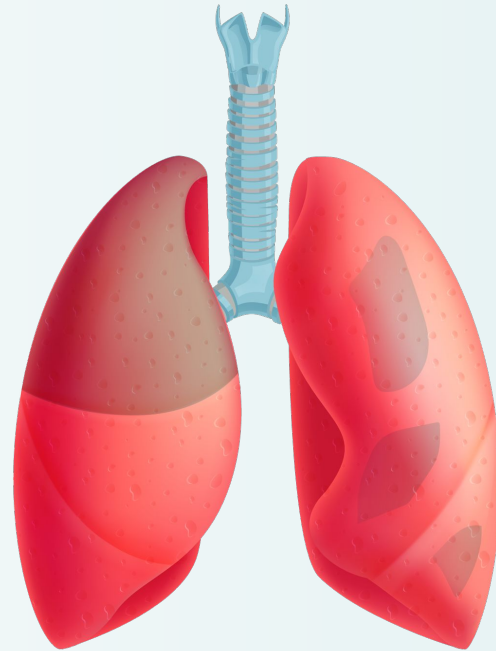
**Breathing/Lung
Anatomy**

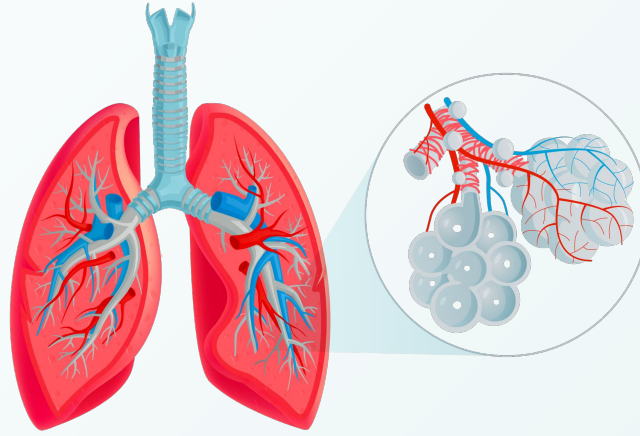
04

Scenario 1

32 YOF with sudden onset of
difficulty breathing

**What is your field
impression?**





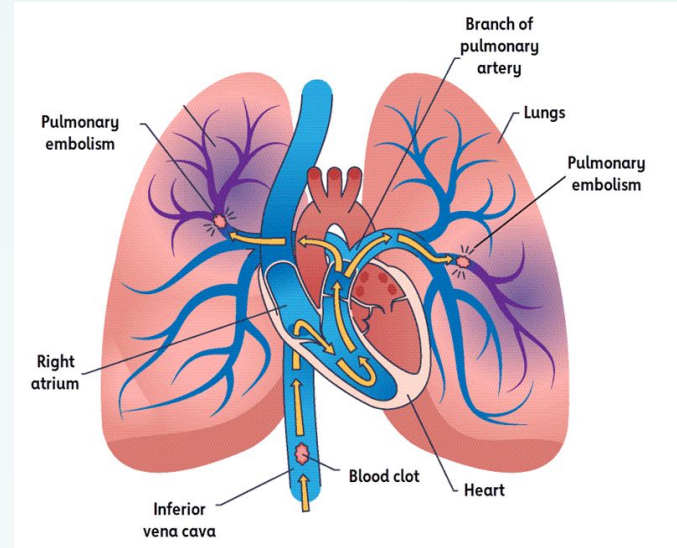
Pulmonary Embolism

What is a PE?

A sudden blockage in an artery in the lung, caused by a blood clot that travels from a deep vein in another part of the body

Causes:

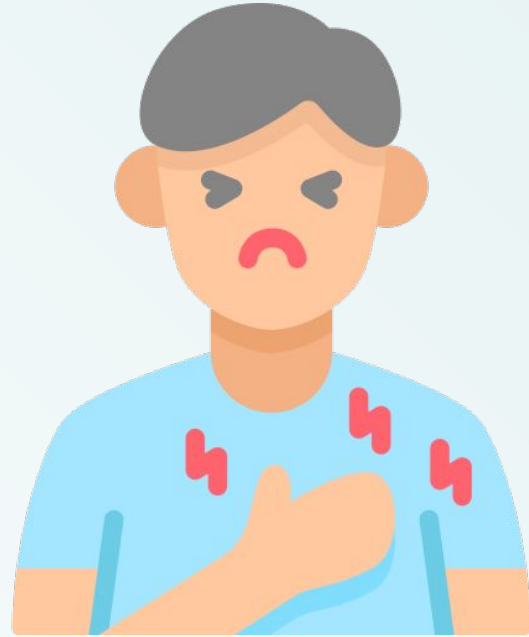
- Deep Vein Thrombosis (DVT)
- Immobility
- Genetic clotting disorders



Signs and Symptoms

- Tachypnea (respirations > 16/min)
- Tachycardia
- Hypotension
- Chest Pain (sharp, worsens when you breathe deeply)
- Dyspnea
- Cough
- Chest tightness

Lung sound are clear bilaterally



Treatment



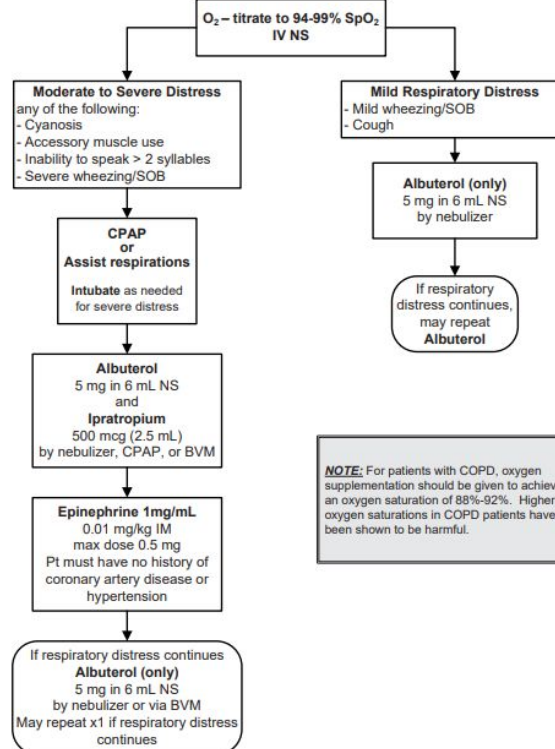
- Rapid Transport
- Transport in position of comfort
 - Allow patient to sit upright if that makes breathing easier
 - Avoid unnecessary exertion
- Oxygen
 - Nasal Cannula: for a patient who is breathing adequately and needs supplemental O₂
 - Non-rebreather: for significant hypoxia with adequate respiratory effort
 - BVM: if respiratory effort become inadequate or the patient becomes apneic
- Reassess every 5 minutes, ALS standby

ALCO Protocol

RESPIRATORY DISTRESS

• Routine Medical Care

- Asthma
- Bronchospasm
- COPD
- Pulmonary edema (see [page 49](#))
- Limit physical exertion, reduce patient anxiety



NOTE: For patients with COPD, oxygen supplementation should be given to achieve an oxygen saturation of 88%-92%. Higher oxygen saturations in COPD patients have been shown to be harmful.

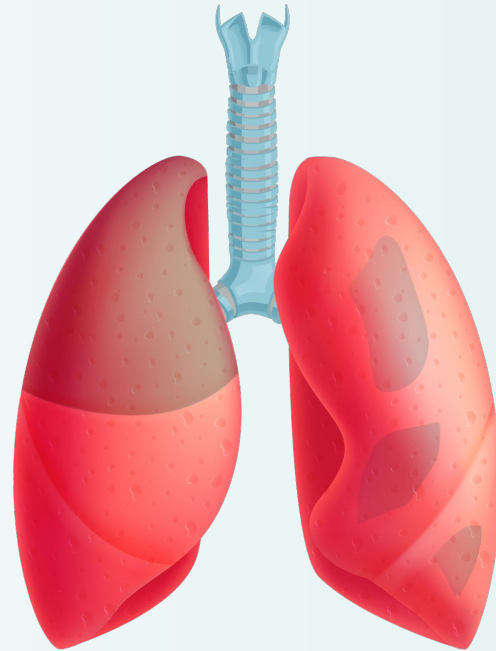
BMRC Protocol

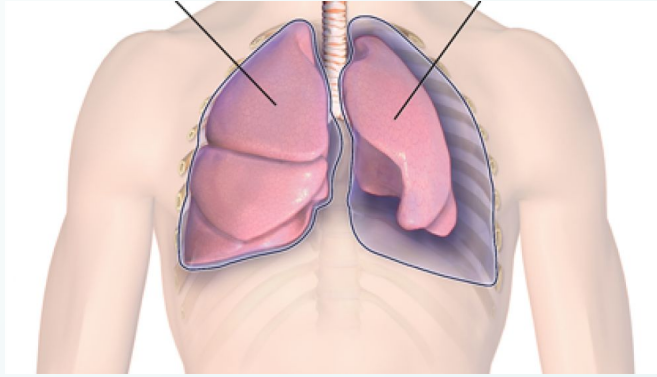
- Obtain OPQRST and SAMPLE
- Place patient in a position of comfort
- Closely monitor vitals every 5 minutes
- Call UCPD or 911
- Be ready to give a report to a higher-level-of-care provider.

Scenario 2

26 YOM with gunshot wound through chest

What is your field impression?





Tension Pneumothorax

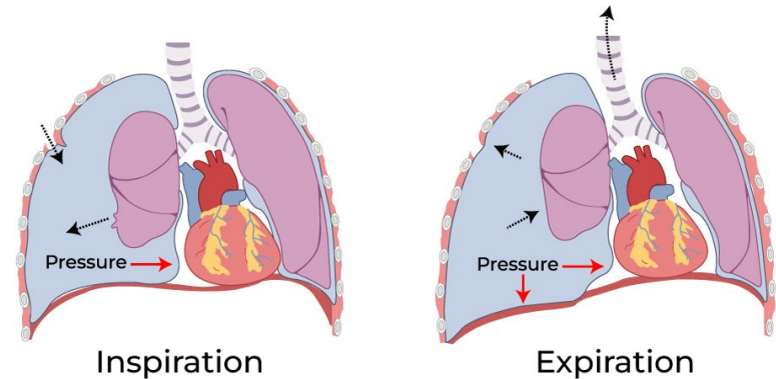
What is a Tension Pneumothorax?

Air enters the pleural space and cannot exit, building intrathoracic pressure. Simple pneumothorax and progress to tension when high pressures crush the superior vena cava.

Causes:

- Penetrating trauma
- Blunt trauma

Tension pneumothorax

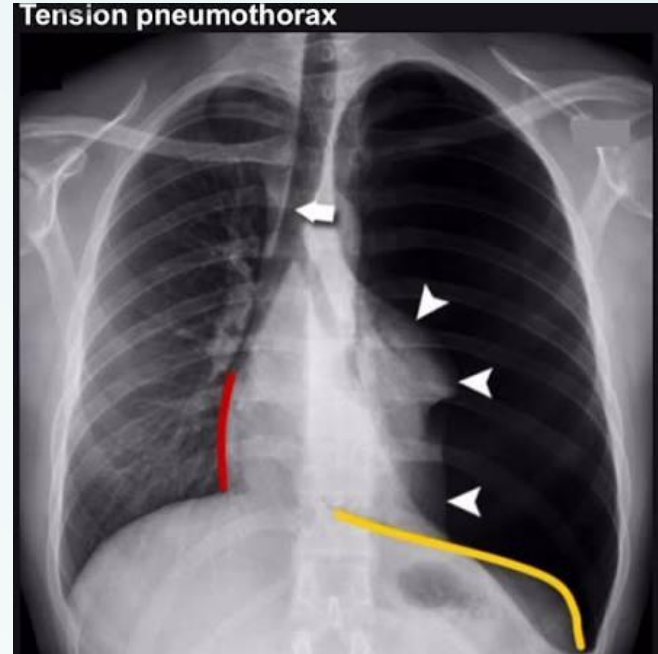


Signs and Symptoms

- Tachypnea (respirations > 16/min)
- Diminished unilateral lung sounds
- Tachycardia
- Hypotension
- Severe Dyspnea
- Low blood oxygen

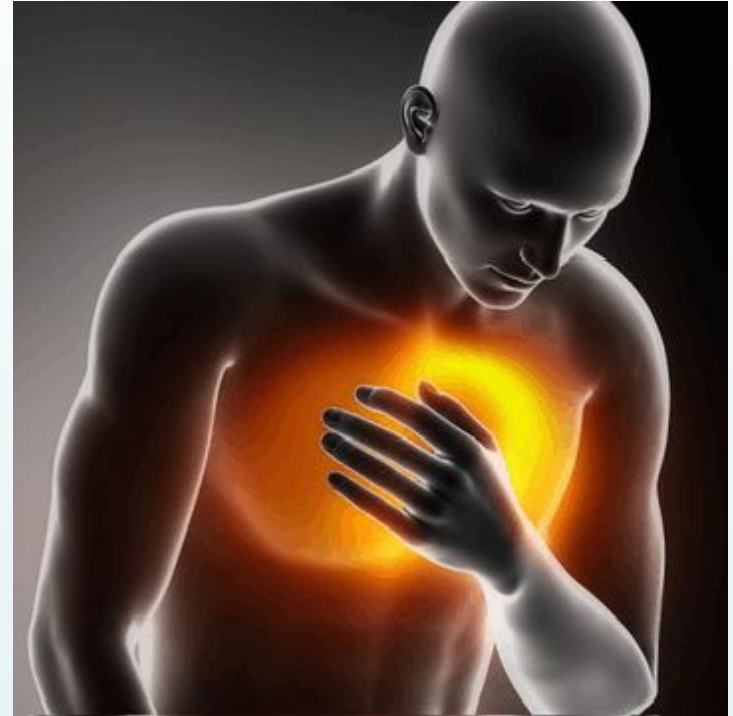
Late signs:

- Tracheal deviation
- JVD
- Cyanosis
- ALOC



Treatment

- Vented chest seal over penetrating chest injuries
- High flow oxygen
 - BVM or NRB
- If patient conscious and can maintain their own airway, place in semi-fowlers if it's easier to breath, or lay supine.
- Keep them warm
- Rapid Transport
- ALS intercept
 - definitive field treatment for a tension pneumothorax is needle thoracostomy, not in our scope

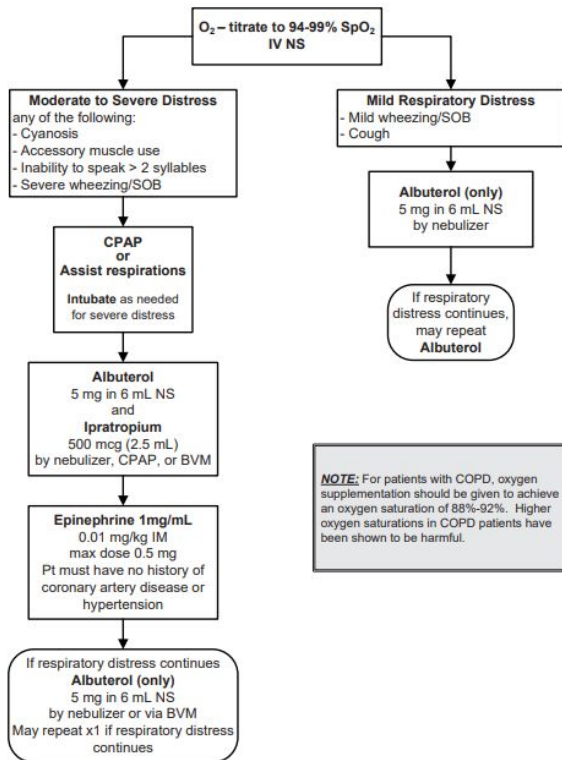


ALCO Protocol

RESPIRATORY DISTRESS

•Routine Medical Care

- Asthma → COPD
- Bronchospasm → Pulmonary edema (see [page 49](#))
- Limit physical exertion, reduce patient anxiety



NOTE: For patients with COPD, oxygen supplementation should be given to achieve an oxygen saturation of 88%-92%. Higher oxygen saturations in COPD patients have been shown to be harmful.

Procedures

Modified On: May 10, 2019

PLEURAL DECOMPRESSION

1. **INDICATIONS:** When clinical findings reveal a tension pneumothorax (severe respiratory distress, diminished breath sounds on the affected side, tracheal deviation) with rapidly deteriorating vital signs
2. **EQUIPMENT:**
 - 2.1 County-approved decompression needle/kit
3. **PROCEDURE:**
 - 3.1 Preferred Site:
 - 2nd or 3rd intercostal space, mid-clavicular line
 - 3.2 Prep site with chlorhexidine
 - 3.3 Firmly but carefully insert the needle at a 90 degree angle just over the superior aspect (superior border) of the rib, through the skin and pleura until air escapes or a distinct "give" is felt. The undersurface of the rib should be avoided to limit injury to the neurovascular bundle. Air should be freely aspirated (if not, you are not in the pleural space)
 - 3.4 Remove the needle
 - 3.5 Attach a one-way valve (if necessary).
 - 3.6 Recheck breath sounds and continuously monitor cardio-respiratory status.
4. **COMPLICATIONS:**
 - 4.1 Lung laceration
 - 4.2 Pneumothorax
 - 4.3 Hemorrhage secondary to damage to the intercostal artery or vein

- CPAP is absolutely contraindicated for signs and symptoms of Pneumothorax

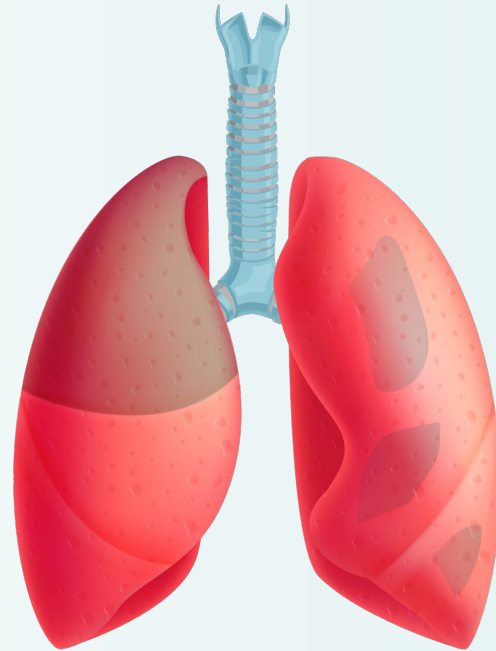
BMRC Protocol

- Call UCPD or 911
- Obtain OPQRST and SAMPLE
- Place patient in a position of comfort
- Closely monitor vitals every 5 minutes
- Be ready to transfer care
- Fill out PCR

Scenario 3

19 YOF with difficulty breathing and wheezing

What is your field impression?



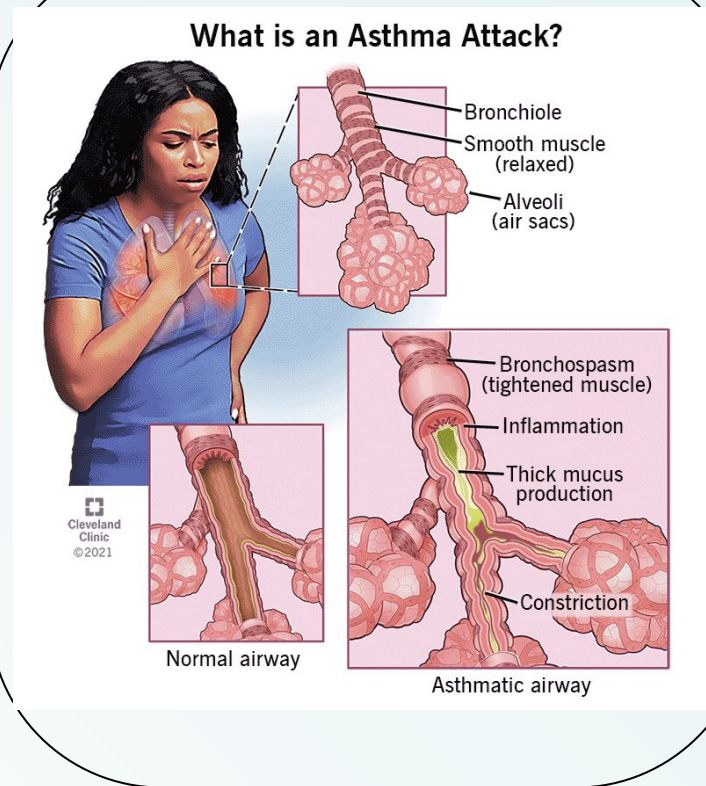


ASTHMA

What is Asthma?

A long-term lung condition that makes your airways swell, tighten, and fill with extra mucus, which causes trouble breathing

Asthma Attack: sudden worsening of asthma symptoms where the airways in your lungs narrow and make it very hard to breathe

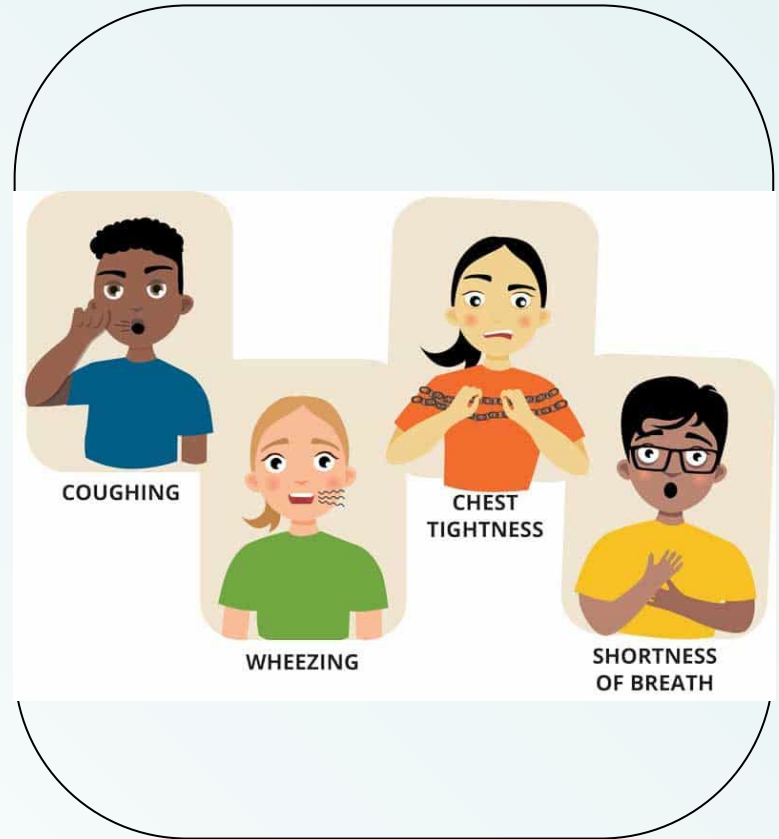


Signs and Symptoms

- Shortness of Breath
- Wheezing
- Chest tightness
- Low blood oxygen
- No improvement even after use of inhaler

Late signs:

- Cyanosis
- ALOC
- Trouble Walking or Talking

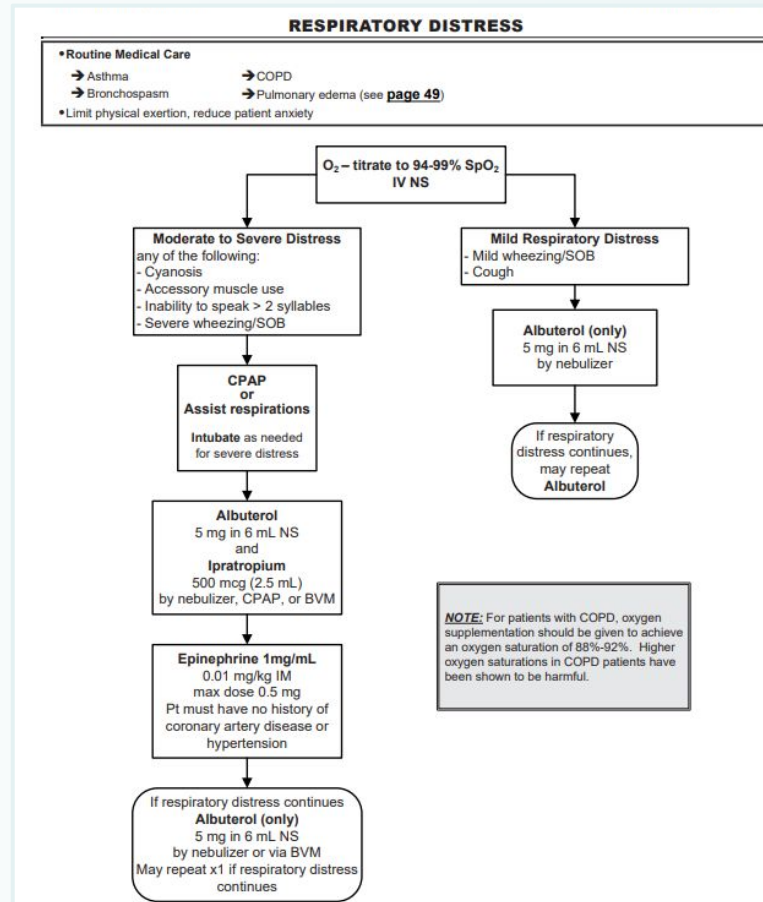


Treatment

- Help patient administer prescribed inhaler
- High flow oxygen
 - BVM or NRB
- Position of comfort
- Rapid Transport
- ALS intercept
 - Especially if patient doesn't have their own prescribed inhaler ALS can administer better and faster bronchodilators which is not in our scope of practice



ALCO Protocol



BMRC Protocol

- Call UCPD or 911
- Obtain OPQRST and SAMPLE
- Place patient in a position of comfort
- Closely monitor vitals every 5 minutes
- Be ready to transfer care
- Fill out PCR

BREATHING PROCESS

Movement of air in and out of lungs (Inhalation and Exhalation)

Pulmonary ventilation

Blood carries oxygen and carbon dioxide throughout the body

Transport of gases

Cells use oxygen to produce ATP, by product carbon dioxide

Cellular Respiration

01

02

03

04

05

External respiration

Exchange of gases between the alveoli and the blood

Internal Respiration

Exchange of gases between the blood and body tissues

ANATOMY

Larynx

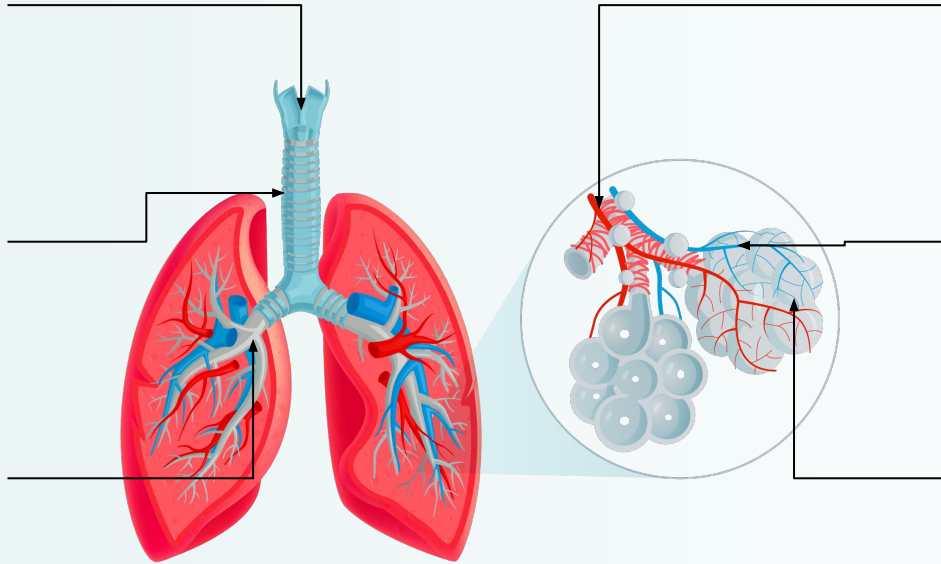
Artery

Trachea

Vein

Bronchi

Capillaries



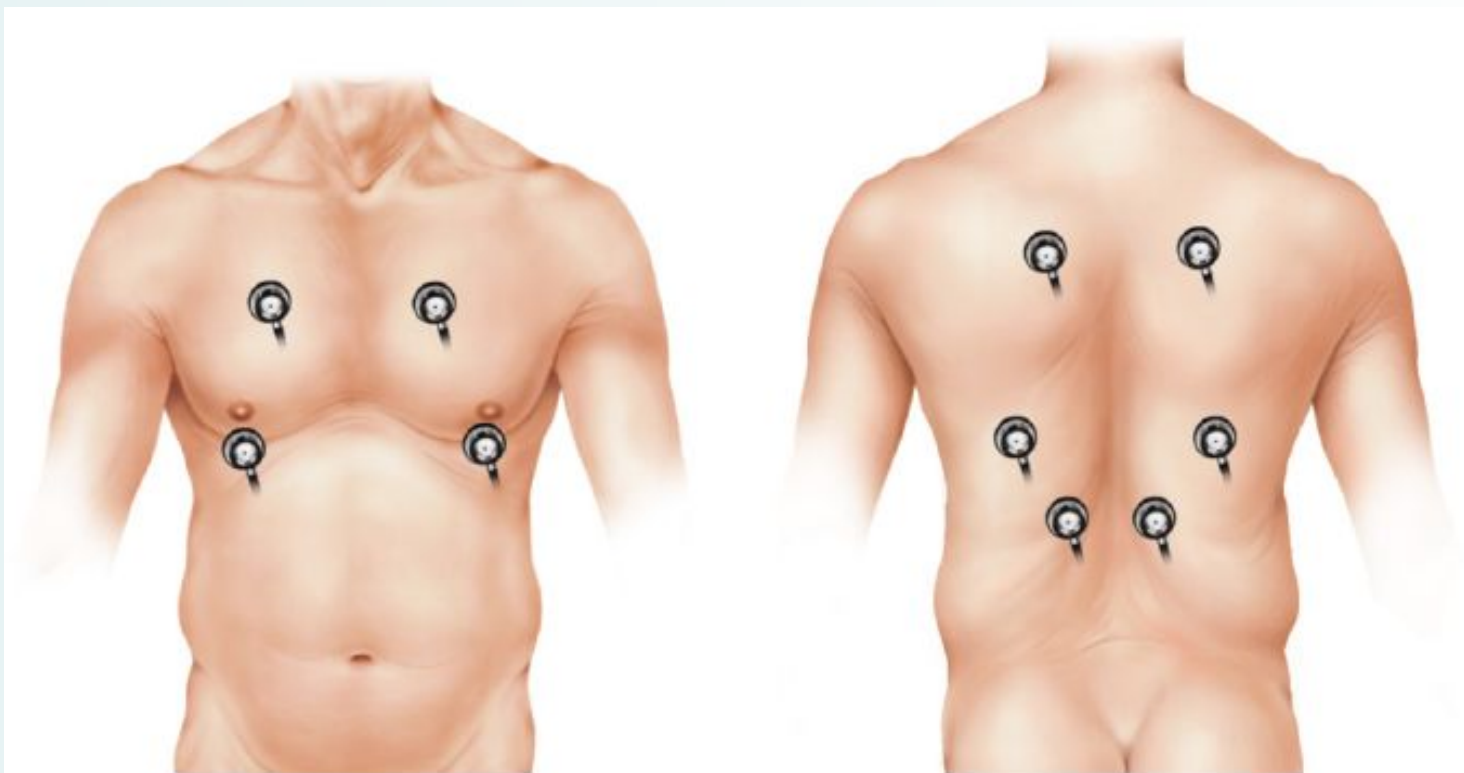


FIGURE 16-15 Locations of the stethoscope bell for auscultation of breath sounds.

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TABLE 16-5 Signs, Symptoms, and Adventitious Breath Sounds Associated With Specific Respiratory Diseases

Breath Sounds	Disease	Signs and Symptoms
Wheezes	Asthma COPD Congestive heart failure/pulmonary edema Pneumonia Bronchitis Anaphylaxis	Dyspnea Productive or nonproductive cough Dependent edema, pink frothy sputum Fever, pleuritic chest pain Clear or white sputum Hives, facial swelling, stridor, nonproductive cough
Rhonchi	COPD Pneumonia Bronchitis	Productive cough Fever, pleuritic chest pain Clear or white sputum
Crackles	Congestive heart failure/pulmonary edema Pneumonia	Dependent edema, pink frothy sputum Fever, pleuritic chest pain
Stridor	Croup Epiglottitis	Fever, barking cough Fever, sore throat, drooling
Decreased or absent breath sounds	Asthma COPD Pneumonia Hemothorax Pneumothorax Atelectasis	Nonproductive cough, dyspnea Productive cough Fever, pleuritic chest pain Shock, respiratory distress Dyspnea, pleuritic chest pain Fever, decreased oxygen saturation

Abbreviation: COPD, chronic obstructive pulmonary disease

THANKS!

Do you have any
questions?

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